NEWTON SCHOOL

ANSWER KEY

Speed, Distance & Time: Word Problems

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Part A: Finding Speed, Distance, or Time (Problems 1–4)

Problem 1

> A car traveled 180 km in 3 hours. Find its average speed in km/h.

Speed Formula

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Speed} = \frac{180\ \text{km}}{3\ \text{h}}$$

$$\text{Speed} = 60\ \text{km/h}$$

Answer: 60 km/h

Problem 2

> A cyclist rides at an average speed of 15 km/h for 4 hours. How far does the cyclist travel?

Distance Formula

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Distance} = 15\ \text{km/h} \times 4\ \text{h}$$

$$\text{Distance} = 60\ \text{km}$$

Answer: 60 km

Problem 3

> A train traveled 240 km at an average speed of 80 km/h. How long did the trip take?

Time Formula

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time} = \frac{240\ \text{km}}{80\ \text{km/h}}$$

$$\text{Time} = 3\ \text{hours}$$

Answer: 3 hours

Problem 4

> David swam 600 m in 15 minutes. Find his average speed in m/min.

Speed Formula

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$
$$\text{Speed} = \frac{600\ \text{m}}{15\ \text{min}}$$
$$\text{Speed} = 40\ \text{m/min}$$

Answer: 40 m/min

Part B: Word Problems (Problems 5–7)

Problem 5

> Jerome walked from his house to the library. His average speed was 50 m/min and it took him 12 minutes. Find the distance from Jerome's house to the library.

Distance Formula

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Distance} = 50\ \text{m/min} \times 12\ \text{min}$$

$$\text{Distance} = 600\ \text{m}$$

Answer: 600 m

Problem 6

> A car traveled from Town A to Town B at an average speed of 60 km/h. The distance between the two towns is 150 km. How long did the trip take?

Time Formula

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time} = \frac{150\ \text{km}}{60\ \text{km/h}}$$

$$\text{Time} = 2.5\ \text{hours}$$

Answer: 2.5 hours (or 2 hours 30 minutes)

Problem 7

> Miguel walked 840 m from home to his school. His average speed was 70 m/min. He left home at 7:30 a.m. At what time did he arrive at school?

Time Formula Clock Time

Step 1: Find time taken

$$\text{Time} = \frac{840\ \text{m}}{70\ \text{m/min}} = 12\ \text{minutes}$$

Step 2: Add to departure time

$$7{:}30\ \text{a.m.} + 12\ \text{min} = 7{:}42\ \text{a.m.}$$

Answer: 7:42 a.m.

Part C: Multi-Part Problems (Problems 8–10)

Problem 8

> Tyler took 2 hours to drive from Town A to Town B, which are 150 km apart. (a) Find his average speed. (b) How long would it take to drive from Town B to Town A if his average speed was 60 km/h?

Speed Formula | Time Formula

(a) Find average speed:

$$\text{Speed} = \frac{150\text{ km}}{2\text{ h}} = 75\text{ km/h}$$

> **(a)** 75 km/h

(b) Find time at 60 km/h:

$$\text{Time} = \frac{150\text{ km}}{60\text{ km/h}} = 2.5\text{ hours}$$

> **(b)** 2.5 hours (or 2 hours 30 minutes)

Answers: (a) 75 km/h | (b) 2.5 hours

Problem 9

> A car traveled from Town X to Town Y (44 km) at 88 km/h. Then from Town Y to Town Z (96 km) at 64 km/h. (a) Find total time taken. (b) Find average speed for whole trip.

Multi-segment | Average Speed

(a) Total time:

Time X to Y: $\frac{44}{88} = 0.5$ hour = 30 minutes

Time Y to Z: $\frac{96}{64} = 1.5$ hours = 1 hour 30 minutes

Total time: $0.5 + 1.5 = 2$ hours

> **(a)** 2 hours

(b) Average speed:

Total distance: $44 + 96 = 140$ km

$$\text{Average speed} = \frac{140\text{ km}}{2\text{ h}} = 70\text{ km/h}$$

> **(b)** 70 km/h

Answers: (a) 2 hours | (b) 70 km/h

Problem 10

> Meihua jogged for $\frac{1}{2}$ hour at 6 km/h. She jogged for another $\frac{1}{4}$ hour at 8 km/h. What distance did she jog altogether?

Distance Formula | Fractions

Distance 1: $\mathbf{6\ km/h \times \frac{1}{2}\ h = 3\ km}$

Distance 2: $\mathbf{8\ km/h \times \frac{1}{4}\ h = 2\ km}$

Total distance: $\mathbf{3 + 2 = 5\ km}$

Answer: 5 km

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